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Frequency Tables

Introduction

It is said that most communication is nonverbal and that a picture contains a thousand words. It can also be said that a table might contain nearly as many words as a picture. Scientists of all fields make extensive use of tables because they are excellent tools with which to communicate large amounts of information in very concise ways. Therefore, it is important to know not only how to read tables but how to create them. In this chapter, we will learn how to read and how to construct frequency tables.

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Chapter Summary

From Chapter 2 of *Statistics and Data Analysis for Social Science*, 1/e. Eric J. Krieg. Copyright © 2012 by Pearson Education. Published by Allyn & Bacon. All rights reserved.

Data Characteristics that can be empirically observed and measured.

Frequency The number of times an event or value occurs.

Frequency table A table for a single variable that indicates the number of cases for each attribute of the variable.

Cross-tabulation table A table that consists of two frequency tables, one in the rows and the other in the columns.

This chapter also marks the point at which *Now You Try It* exercises are included. You will see these appear in strategic places throughout this chapter. They are intended to provide students with opportunities to test their comprehension of the main concepts that are presented in the chapter. Answers to the *Now You Try It* exercises are found at the end of each chapter.

Before we begin, Table 1 contains a few important symbols and formulas that are used.

TABLE 1 *Symbols and Formulas*

SYMBOL	MEANING OF SYMBOL	FORMULA
N	The number of cases in a table	
F	The frequency of cases for a particular attribute	
Cf	The cumulative frequency of cases for a group of attributes	Σf
P	The sample proportion	$P = \frac{f}{N}$
π	The population proportion	$\pi = \frac{f}{N}$
%	Percent	$\% = \frac{f}{N}(100)$
$c\%$	Cumulative percent	$c\% = \frac{cf}{N}(100)$
Ratio	The relative frequency of cases across different populations	$Ratio = \frac{f_1}{f_2}$

Toxic Waste and “Superfund”

In 1980, President Ronald Reagan signed into law the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), otherwise known as “Superfund.” It is considered by many to be the most ambitious piece of environmental legislation ever written and, among other goals, was intended to help fund the cleanup of the nation’s worst hazardous waste sites.

The passage of Superfund came about when national news coverage of a neighborhood in Niagara Falls, New York, alerted the public to the possibility of toxic wastes being present in their communities. This neighborhood, Love Canal, is a typical blue-collar working-class community that in many ways could be considered “Anywhere USA.” As sociologist Andrew Szasz (1994) argues, mass media brought the issue into the homes of nearly all Americans. The news coverage was the result of the work of a citizen organization, the Niagara Falls Homeowners Association (NFHA), led by a woman named Lois Gibbs. Gibbs and some of her neighbors felt that their community contained unusually high numbers of miscarriages, stillbirths (babies born dead), and children with birth defects. Watching community activism unfold to reveal the extent of the toxic threat, millions of Americans were left



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Figure 1 Love Canal Neighborhood
Before and After the Toxic Waste Disaster

with the feeling that if the Love Canal community was contaminated, almost any community could be contaminated.

Eventually, the federal government was persuaded to offer the residents of Love Canal a “buyout” option in which homeowners would be offered a “fair” market value for their home. Most residents accepted the buyout as a way to escape the toxic threats to their health; however, many noted that while their home ownership investments were saved, their community was ripped apart and lost.

Although many other communities around the country are well known for their fight to protect themselves (Times Beach, Missouri, and Warren County, North Carolina), Love Canal is credited as a landmark case that spurred the passage of CERCLA in 1980. Superfund, as CERCLA is commonly referred to, consists of a legal and financial plan to clean up the nation’s most contaminated toxic sites and deter companies from illegally dumping waste.

The “fund” in Superfund was originally a \$1.6 billion trust fund to be used (1) for litigation against responsible parties and (2) for cleanup efforts when responsible parties could not be identified. In addition to holding polluters responsible for their wastes, Superfund imposes \$10,000 per-day fines against responsible parties that do not come forward to claim their responsibility. To help ensure that they do come forward, Superfund can hold multiple parties responsible. Overall, the attempt was to make it no longer pay to pollute.

It is debatable how successful these efforts have proven. Certainly, a great deal more cleanup has been done with Superfund than would have been achieved without it. And because of the Superfund Amendment and Reauthorization Act of 1986 that added an additional \$5.8 billion to the Superfund, citizens now have the rights and the tools to find out what chemicals are being used and disposed of in their communities.

Approximately 1,200–1,300 active Superfund sites are scattered across the country, with tens of millions of Americans living within a few miles of them. Hundreds of other Superfund sites have been cleaned up. Despite this alarming number of hazards, it is likely that tens or hundreds of thousands of other, less dangerous sites exist across the country.

In fact, by the late 1980s, a number of scientific studies revealed that toxic waste sites are far more common than people previously believed. Superfund sites are but one type of hazard and scientific investigations soon showed that other, less dangerous sites were quite common in most communities. Leaking underground storage tanks, toxic spills,

industrial emissions, waste transfer stations, and a variety of other ecological hazards pose a greater risk to overall public health than do Superfund sites listed on the Environmental Protection Agency's *National Priority List* (NPL). Nevertheless, as of May 2, 2008, the NPL contained 1,641 sites in the United States, Guam, Puerto Rico, U.S. Virgin Islands, Trust Territories, American Samoa, and the Northern Mariana Islands.

Integrating Technology

If you would like to conduct research on your community, try visiting the Environmental Protection Agency's *TRIEXPLORER* (<http://www.epa.gov/triexplorer/>). This website includes a search engine that enables users to view many of the ecological hazards that may be present in your community, including the number of pounds of toxic wastes released directly into the environment since 1986, the number of National Priority List (Superfund) sites, and others. It allows users to identify which companies are responsible for toxic releases, how much was released, and how they were released (to the air, water, soil, or transferred off-site).

Individual data Data that represents characteristics of individuals (people, houses, cars, dogs, etc.).

Ecological data Data that represents characteristics of groups (towns, cities, counties, etc.).

Individual and Ecological Data

Now is a good time to recap the difference between individual data and ecological data. Individual data is representative of a single person. For example, if we conducted a sample of college students our unit of analysis is individual students. Ecological data, on the other hand, is representative of groups or areas. For example, the number of police cars in each town is a characteristic of the towns we are studying, not the individuals in the towns. The data generated in the U.S. Census is probably one of the best examples of ecological data. Data from the census can be used to generate thousands of statistics at the national, state, county, town, and even block levels; however, it is impossible to use census data to learn anything about any one individual. In this sense, census data is ecological data because it is representative of geographic areas and not of individuals.

Frequency Tables

Frequency tables allow us to organize large quantities of data so that they can be described and communicated with others easily by summarizing the distribution of cases across the attributes of a single variable. Typically, frequency tables include more than just the frequencies of cases; they also include a variety of percentages. This chapter introduces readers to the role that frequencies, proportions, percentages, and cumulative percentages play in the use of tables. We look at how Superfund sites are classified and how they are geographically distributed nationwide by state. Before doing so, however, we first analyze some data that is representative of individuals (as opposed to Superfund sites or states).

Before we begin our discussion, let's take a moment to see what a frequency table actually looks like. Table 2 is a frequency table for the variable SEX. SEX is operationalized as Male or Female. As you can see, the table provides a great deal of information, including the frequency of males ($f = 2,003$) and the frequency of females ($f = 2,507$). Together, these add up to the number of respondents ($N = 4,510$).

Often, some respondents are unwilling or unable to provide data. In these cases, they are often counted as "missing data." They are typically shown in the frequency column below the total. Table 2, a graphic representation of which is shown in

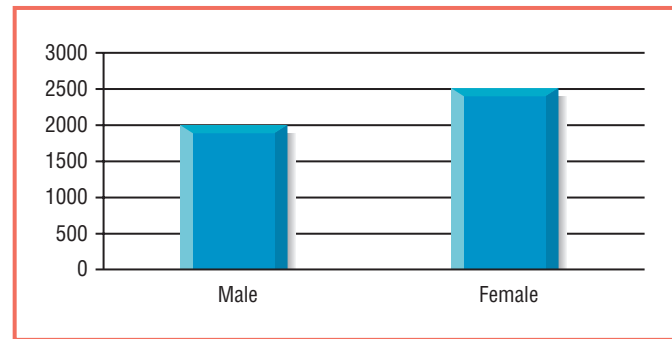
TABLE 2 *Respondent's Sex*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	MALE	2003	44.4	44.4	44.4
	FEMALE	2507	55.6	55.6	100.0
	Total	4510	100.0	100.0	

Figure 2, does not have any missing values in it, but tables discussed later in the chapter do. It is important to realize that missing data are not included in any statistics.

Proportions, Percentages, and Ratios

Proportions. It is important to understand the basics behind proportions and percentages before working with frequency tables. Proportions and percentages allow us to describe groups of cases, and allow us to make comparisons across populations.

**Figure 2** Respondent's Sex

EXAMPLE 1

Satisfaction with Social Life

Suppose we sample 500 males and 600 females. Of the 500 males, 375 are satisfied with their social life. Of the 600 females, 425 are satisfied with their social life.

Based on these numbers, it is difficult to determine whether males or females tend to be more satisfied with their social life. Proportions and percentages allow us to overcome this problem. We can mathematically represent 375 out of 500 males being satisfied with their social life with the following formula:

$$P = \frac{f}{N}$$

In this formula, f is equal to the number of males who indicate they are satisfied with their social life and N is equal to the total number of males who answered the question. Therefore:

$$P = \frac{375 \text{ satisfied}}{500 \text{ total}} = \frac{375}{500} = .750$$

Proportion A way to standardize the frequency of cases so that comparisons can be made across populations.



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Therefore, the proportion (P) of males who are satisfied with their social life is .750.

Using the same formula for females, we find that:

$$P = \frac{f}{N} = \frac{425 \text{ satisfied}}{600 \text{ total}} = \frac{425}{600} = .708$$

Based on this we can compare the proportion of males who are satisfied to the proportion of females who are satisfied and draw a conclusion as to which group is most likely to be satisfied with their social life. In this case, the answer is males because they have a higher proportion.

All proportions range between a low of 0 and a high of 1.0. This is because the value of the denominator in the proportion equation is always greater than or equal to the value of the numerator. In the calculation of a proportion, the numerator is never larger than the denominator.

Percentage A way to standardize the frequency of cases as the number of responses per 100 cases.

Percentages. Percentages are extremely easy to calculate once you know how to calculate proportions. To determine a percentage, first, calculate a proportion and, second, multiply the proportion by 100. That's it! The formula for a percentage is shown below:

$$\% = \frac{f}{N}(100)$$

Using our example above, we can calculate the percent of males who are satisfied with their social life as follows:

$$\% = \frac{f}{N}(100) = \frac{375}{500}(100) = .75(100) = 75.0$$

That is, 75% of males are satisfied with their social life.

Similarly, for females:

$$\% = \frac{f}{N}(100) = \frac{425}{600}(100) = .708(100) = 70.8$$

As with proportions, we can now compare males to females. While 75.0% of males are satisfied with their social life, only 70.8% of females are satisfied.

Ratios. A ratio is another important statistical tool that allows us to communicate more easily. Using our example above, suppose someone asked us “How many females are in your sample for each male?” In other words, they want to know the ratio of females to males. We could say that for each 600 females we have 500 males, but this is not easy to understand. The solution is fairly simple using the formula below:

$$\text{Ratio} = \frac{f_1}{f_2}$$

In this case, f_1 is equal to the number of females and f_2 is equal to the number of males. Essentially, we are standardizing the number of males as 1 by putting the frequency for males in the denominator of our equation.

$$\text{Ratio} = \frac{f_1}{f_2} = \frac{600}{500} = 1.2$$

This tells us that for each male, there are 1.2 females.

A way to remember which frequency goes in the numerator and which goes in the denominator is to use the phrasing of the question being asked. For example, if we want to know the ratio of females to males, we put the frequency for males in the denominator. If we want to know the ratio of males to females, we put the frequency of females in the denominator. Essentially, whichever variable follows the word *to* in our question is the one that goes in the denominator.

Ratio A way to compare the relative frequency of cases across populations.

Now You Try It 1

Suppose we want to know about students’ attitudes toward a ban on tobacco use on campus. We decide to survey 500 students, 355 of whom live on-campus and 145 of whom live off-campus. Of those who live on-campus, 276 indicate that they approve of a tobacco ban and of those who live off-campus, 92 approve of a tobacco ban.

Use this information to answer the following questions.

1. What proportion of all students approve of a tobacco ban?
2. What percent of on-campus students approve of a tobacco ban?
3. What percent of off-campus students approve of a tobacco ban?
4. What is the ratio of on-campus to off-campus students?

*Answers at end of chapter.

EXAMPLE 2

Public Opinion on Environmental Spending

How do people in the United States view the environment? Do they feel we are doing enough to protect it? These questions are easy to ask but difficult to answer.

Americans have an interesting relationship with their environments. Generally, the U.S. population subscribes to a view of nature known as “Western dualism.” As the word “dualism” implies, it is really two views of nature that are held simultaneously. On the

Value labels Descriptive labels for the attributes of a variable.

Frequency A column in a frequency table that shows the number of times a particular attribute occurs.

Percent A column in a frequency table that standardizes frequencies by expressing them as the number of times an attribute occurs per 100 cases. Based on all cases.

Valid percent A column in a frequency table that standardizes frequencies by expressing them as the number of times an attribute occurs per 100 cases. Based on only those cases that provided data.

Cumulative percent A column in a frequency table that shows the percent of cases above or below a certain point or attribute.

one hand, we tend to think of nature as an object—something that is “out there,” separate from us, and to be used to make our lives better (think of burning coal for electricity). On the other hand, we tend to think of nature as an extension of ourselves—a part of who we are that should be protected for its own sake. Although most of us probably lean toward one or the other of these two views, it is likely that we have a little of both in us.

Data from the 2006 General Social Survey shed light on which of these two views tends to dominate the American mindset. They represent varying opinions on whether we are spending too little, about right, or too much on improving environmental protections (Figure 3). We might claim that those claiming that we spend too little to protect the environment tend to feel that nature has intrinsic value and those claiming that we spend too much tend to objectify nature.

Table 3 provides us with a great deal of information on the variable itself and the distribution of cases. As you can probably tell, the variable is ordinal because the attributes can be rank-ordered from high to low (from “too little” to “too much”). The table consists of a series of five columns: (1) Value Labels, (2) Frequencies, (3) Percents, (4) Valid Percents, and (5) Cumulative Percents.

Table 3 also tells us that the sample consisted of 4,510 respondents; however, it is important to note that 3,064 of those respondents are classified as “missing.” This means that, for whatever reasons, either the question did not apply to them (NAP) or they did not provide an answer to the question (DK). After removing the 3,064 missing cases (3,026 NAP cases and 38 DK cases), we are left with 1,446 cases. Therefore, we really have two values of N , but for all intents and purposes missing cases are almost always excluded from the analysis. Consequently, $N = 1,446$.

The Frequency column in the table tells us that 992 respondents indicated that they feel the government is spending too little on protecting the environment. The Percent column tells us that these 992 respondents make up 22.0% of the 4,510 included in the sample. The Valid Percent column tells us that these 992 respondents make up 68.6% of the 1,446 respondents who provided data. Each row in the table is read the same way.

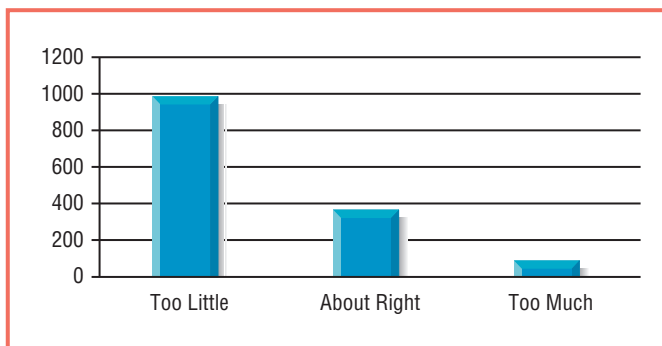


Figure 3 Improving Protecting Environment

TABLE 3 *Improving Protecting Environment*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	TOO LITTLE	992	22.0	68.6	68.6
	ABOUT RIGHT	365	8.1	25.2	93.8
	TOO MUCH	89	2.0	6.2	100.0
	Total	1446	32.1	100.0	
Missing	NAP	3026	67.1		
	DK	38	.8		
	Total	3064	67.9		
Total		4510	100.0		

The last column in the table, the Cumulative Percent column, is read slightly differently. For example, the table tells us that 365 respondents felt that we are spending about the right amount on environmental protection, which happens to be 25.2% of respondents who provided data. The 93.8% in the Cumulative Percent column is based on the total number of respondents who felt that we are spending either too little or about right. In other words, the 93.8% is calculated by adding the 992 who answered too little with the 365 who answered about right. This means that 1,357 respondents answered either too little or about right and these 1,357 respondents comprise 93.8% of the 1,446 valid responses.

It is interesting that despite 68.6% of the American public claiming that we spend too little on environmental protection, we have such a staggering array and degree of environmental problems that have yet to be solved. We now take a look at some Superfund data to see how it can be organized into frequency tables and how to calculate the various percent column values.

Superfund Sites: Frequency Table for a Nominal Variable

Our unit of analysis is the waste site and all Superfund waste sites are listed on what is called the *National Priorities List*. Not all of the sites on the list are currently active. Some sites are listed as *active/final* while others are listed as *deleted* or *proposed*. The difference is that not all sites have undergone sufficient evaluation by the Environmental Protection Agency (EPA) to qualify for the Superfund Program. Those that are still under review are considered *proposed* sites. Those that are either cleaned or are not deemed dangerous enough to meet Superfund requirements are considered *deleted*. And those that are deemed worthy of Superfund actions are considered *active/final*.

Table 4 was taken from the website scorecard.org on November 12, 2008, and ranks each state by the number of Superfund sites that were considered either “active/final” or “proposed” between 1993 and 2004. You can use the website to help assess the level of toxic contamination in your community.

Table 4 lists only sites that are either final or proposed and does not include all Superfund sites. A more complete list also includes those that are listed as *deleted*. Generally, deleted sites are those that have been cleaned up. Including deleted sites gives us a better sense of the total hazards that a state has faced over time.

Using data collected directly from the Environmental Protection Agency website on November 13, 2008, we see that each of the 1,650 sites falls into one and only



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TABLE 4 *Superfund Sites by State*

RANK	STATE	NUMBER OF SUPERFUND SITES
1.	NEW JERSEY	116
2.	CALIFORNIA	98
3.	PENNSYLVANIA	95
4.	NEW YORK	93
5.	MICHIGAN	69
6.	FLORIDA	52
7.	WASHINGTON	47
8.	ILLINOIS	45
	TEXAS	45
9.	WISCONSIN	40
10.	OHIO	35
11.	MASSACHUSETTS	32
12.	INDIANA	30
	VIRGINIA	30
13.	NORTH CAROLINA	29
14.	MISSOURI	27
15.	SOUTH CAROLINA	25
16.	MINNESOTA	24
17.	NEW HAMPSHIRE	20
18.	MARYLAND	19
	UTAH	19
19.	COLORADO	18
20.	CONNECTICUT	16
	GEORGIA	16
	LOUISIANA	16
21.	ALABAMA	15
	DELAWARE	15
	MONTANA	15
22.	IOWA	14
	KENTUCKY	14
23.	KANSAS	13
	NEW MEXICO	13
	TENNESSEE	13
24.	MAINE	12
	OREGON	12
	RHODE ISLAND	12
25.	ARKANSAS	11
	NEBRASKA	11

FREQUENCY TABLES

RANK	STATE	NUMBER OF SUPERFUND SITES
26.	OKLAHOMA	11
	PUERTO RICO	10
	VERMONT	10
27.	ARIZONA	9
	IDAHO	9
	WEST VIRGINIA	9
	ALASKA	6
28.	MISSISSIPPI	5
30.	HAWAII	3
31.	GUAM	2
	SOUTH DAKOTA	2
	VIRGIN ISLANDS	2
	WYOMING	2
32.	DISTRICT OF COLUMBIA	1
	NEVADA	1

There is limited information about many potentially significant sources of contamination. Scorecard's profiles of hazardous waste sites are limited to active/final and proposed sites on the National Priority List and are derived from multiple sources dating from 1993 to 2004.

one of these three attributes. Therefore, the variable is operationalized in a way that its attributes are both collectively exhaustive and mutually exclusive. By counting the number of sites in each attribute, we can quickly and easily describe the distribution of Superfund sites in Table 5.

As you can see in the bar chart in Figure 4, the height of each bar corresponds to its frequency (or valid percent). Bar charts are good ways of displaying data at the nominal or ordinal levels. The data labels at the top of each bar can represent frequencies, percentages, or both depending on what you would like to display. In this case, they show frequencies. The same data are shown in the form of a pie chart in Figure 5. The relative size of each pie slice represents either the number of cases or the valid percent of cases.

TABLE 5 *National Priority List Status*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Deleted	330	20.0	20.0	20.0
	Active/Final	1257	76.2	76.2	96.2
	Proposed	63	3.8	3.8	100.0
	Total	1650	100.0	100.0	

FREQUENCY TABLES

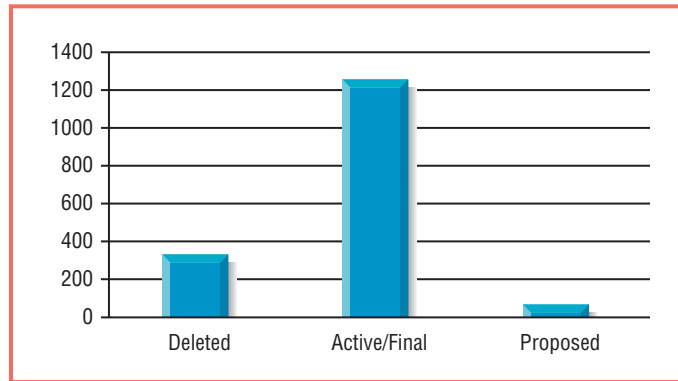


Figure 4 National Priority List Status

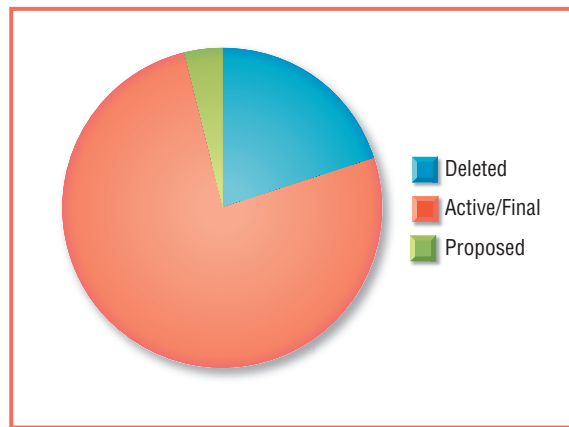


Figure 5 National Priority List Status

Site Status is operationalized as a nominal variable because the attributes cannot be rank-ordered from high to low. It is often tempting to think of some nominal variables as ordinal variables. In the case of “Site Status,” for example, some might claim that we can rank the attributes because active/final sites pose a greater threat than do deleted or proposed sites. Two problems arise with this kind of logic. First, a proposed site could actually pose a much greater threat than an active/final site. Second, the variable is not intended to assess the risk posed by a site; it is only intended to describe how the EPA has classified the site. If we wanted to rank sites according to the threats that they pose, we would have to create new variables with new attributes based on new concepts and measurements. As it is, no ranking can be done and the variable must be treated as nominal.

The frequency table provides a significant amount of information. First, it tells us how many cases are included in the table, N . In this case, N is equal to 1,650. Second, it tells us the frequency of each attribute (hence the name “frequency table”). For example, of the 1,650 sites, 330 are labeled Deleted from the list. Likewise, it tells us that 1,257 sites are Active/Final and 63 are Proposed. Third, it tells us the percent of the time that each attribute occurs. For example, the 330 Deleted sites constitute 20.0% of the 1,650 total sites. We now look at how the percent, valid percent, and cumulative percent are computed.

EXAMPLE 3**Calculating Proportions and Percents Using Superfund Site Data**

Before we can calculate a percent, we must first calculate a proportion. Proportions are always expressed in decimal form and range between 0 and 1.0. Proportions can be turned into percentages by multiplying them by 100 (essentially just moving the decimal two places to the right). The formulas for proportion (P), percent (%), and cumulative percent ($c\%$) are shown below. It should be noted that the percent and valid percent are not always the same, as they are in the case of “Site Status.” They are based on different values of N . The percent is always based on the total number of cases in the table whereas the valid percent is based on only those cases that actually provide data. It is almost always preferable to use the valid percent and not the percent.

Formula for calculating the proportion: $P = \frac{f}{N}$

Formula for calculating the percent: $\% = \frac{f}{N}(100)$

Formula for calculating the cumulative percent: $c\% = \frac{cf}{N}(100)$

We begin by calculating proportions for each of the three attributes. These are computed by taking the frequency of sites for a particular attribute and dividing by the total number of cases.

$$P = \frac{330}{1650} = .200 \quad .200(100) = 20.0\%$$

$$P = \frac{1257}{1650} = .762 \quad .762(100) = 76.2\%$$

$$P = \frac{63}{1650} = .038 \quad .038(100) = 3.8\%$$

By dividing the number of cases in each attribute (f) by the total number of cases in the table (N), we obtain a proportion. By multiplying the proportion by 100, we obtain a percent. For example, the proportion of cases that are deleted is .200 and the percent of cases that are deleted is 20.0%.

You can see that next to the Percent column in Table 5 is a Valid Percent column. In this particular table, the two columns are the same; however, this is not always the case. Oftentimes, frequency tables contain what are called “missing” cases. An example of a table with missing cases is presented later.

The final column in the frequency table is called the Cumulative Percent column. Cumulative Percent columns offer a running total of the Frequency column. For example, you can see that the first cumulative percent value is 20.0%. The next is 96.2%. This value is based on a cumulative frequency (cf) and is obtained by adding the frequency for deleted to the frequency for active, dividing by N , and multiplying by 100. For example:

$$c\% = \frac{330 + 1257}{1650}(100) = 96.2\%$$

This means that 96.2% of all Superfund sites are either Deleted or Active/Final.

Now You Try It 2

Suppose you are doing a study of student integration into social life on campus and want to find out what percent of students live on campus. Using a survey research method, you ask students about their sex, class standing, and what percent of their time they spend on campus. You then test out your questionnaire on a group of students in a dining hall and find the following:

RESPONDENT #	SEX	CLASS STANDING	TIME ON CAMPUS
1	Male	Freshman	90
2	Male	Junior	75
3	Female	Junior	60
4	Female	Sophomore	85
5	Female	Freshman	95
6	Male	Senior	50
7	Female	Junior	70
8	Female	Senior	60
9	Male	Sophomore	90
10	Female	Junior	70

Using the data from the test of your questionnaire, construct a frequency table for the variable Sex. Be sure to include value labels, frequencies, percents, valid percents, and cumulative percents. It is also standard procedure to include a title with each table.

*Answers at end of chapter.

Frequency Table for an Ordinal Variable

For this example the NPL sites are organized into four groups based on the number of sites in each state. Our unit of analysis is no longer the waste site. The unit of analysis is now the state because we are looking at characteristics of states, not characteristics of waste sites. Therefore, the data presented here is ecological data. Territories were removed from the data and only the 50 U.S. states and the District of Columbia are included. This reduces the overall number of cases to 51, so that $N = 51$. It also reduced the number of sites in these states from 1,650 to 1,616 because some geographic regions have been left out of the analysis.

EXAMPLE 4

Analyzing Superfund Site Data by State

The states are now organized into four groups (attributes) based on the number of waste sites. The first group of states consists of those that contain anywhere from 0 to 15 Superfund sites. The second group of states consists of those that contain anywhere from 16 to 30

TABLE 6 *Frequency of Superfund Sites*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	0–15 sites	18	35.3	35.3	35.3
	16–30 sites	16	31.4	31.4	66.7
	31–45 sites	7	13.7	13.7	80.4
	46 or more sites	10	19.6	19.6	100.0
Total		51	100.0	100.0	

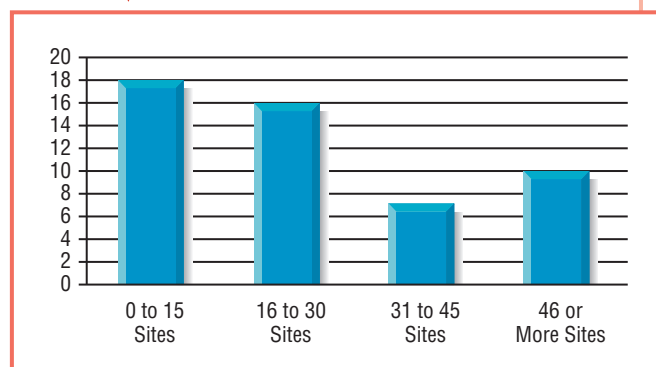
Superfund sites. The third group of states consists of those that contain anywhere from 31 to 45 Superfund sites. The final group of states consists of those that contain 46 or more sites. Table 6 shows the distribution and Figure 6 shows the distribution in the form of a bar chart.

Table 6 indicates that 18 states have between 0 and 15 sites; 16 states have between 16 and 30 sites; 7 states have between 31 and 45 sites; and 10 states have 46 or more sites. The attributes are mutually exclusive and collectively exhaustive. As you can see, all of the columns in the frequency table are the same as they were for the previous one. They are the same for all frequency tables generated by SPSS for Windows.

This variable is ordinal, although it is easy to be fooled into thinking that it is interval/ratio because the attributes are represented as ranges of numbers. Here is the justification. When looking at the attributes, it is possible to say that those states with 31–45 sites have more than those states with 16–30 sites. On the other hand, we cannot tell exactly how many more sites a state with “16–30” sites contains relative to a state with “0–15” sites. It could be one additional site or it could be 30 additional sites. Or it could be anything in between. The point is we just don’t know; nor can we tell from the table. Therefore, we can only say that a particular state has more or less than another state, thereby making the variable ordinal. Because we are unable to determine the exact difference in the frequency of sites between the two states, the variable must be ordinal.

This is called grouped data and it is important to realize that just because the table has numbers for the attributes, it does not make it an interval/ratio variable.

All of the percentages are calculated using the same methods and formulas that were used to calculate them in the previous table using the formula for percentage (%).

Figure 6 Frequency of Superfund Sites

FREQUENCY TABLES

$$\% = \frac{f}{N}(100)$$

0–15 sites: $\% = \frac{18}{51}(100) = 35.3$ Therefore, 35.3% of all states contain 0–15 Superfund sites.

16–30 sites: $\% = \frac{16}{51}(100) = 31.4$ Therefore, 31.4% of all states contain 16–30 Superfund sites.

31–45 sites: $\% = \frac{7}{51}(100) = 13.7$ Therefore, 13.7% of all states contain 31–45 Superfund sites.

46 or more sites: $\% = \frac{10}{51}(100) = 19.6$ Therefore, 19.6% of all states contain 46 or more Superfund sites.

The Cumulative Percent column is more applicable for tables with ordinal variables. Cumulative percents are calculated using the following formula:

$$c\% = \frac{cf}{N}(100)$$

0–15 sites: $c\% = \frac{18}{51}(100) = 35.3$ Therefore, 35.3% of all states contain 0–15 Superfund sites.

0–30 sites: $c\% = \frac{18 + 16}{51}(100) = 66.7$ Therefore, 66.7% of all states contain 0–30 Superfund sites.

0–45 sites: $c\% = \frac{18 + 16 + 7}{51}(100) = 80.4$ Therefore, 80.4% of all states contain 0–45 Superfund sites.

0–46 or more sites: $c\% = \frac{18 + 16 + 7 + 10}{51}(100) = 100.0$ Therefore, 100.0% of all states contain 0–46 or more Superfund sites.

Now You Try It 3

Using the preliminary data from your study of integration into campus social life, construct a frequency table for the variable Class Standing. Be sure to include value labels, frequencies, percents, valid percents, and cumulative percents.

RESPONDENT #	SEX	CLASS STANDING	TIME ON CAMPUS
1	Male	Freshman	90
2	Male	Junior	75
3	Female	Junior	60
4	Female	Sophomore	85
5	Female	Freshman	95
6	Male	Senior	50

RESPONDENT #	SEX	CLASS STANDING	TIME ON CAMPUS
7	Female	Junior	70
8	Female	Senior	60
9	Male	Sophomore	90
10	Female	Junior	70

*Answers at end of chapter.

Frequency Table for an Interval/Ratio Variable

For this example the NPL sites are organized by states and the District of Columbia (hence $N = 51$ instead of 50). Unlike the last example the states are not grouped into categories on the basis of how many sites they contain. Yet we use the same formula for percent (%) and cumulative percent that we used for the previous tables.

$$\% = \frac{f}{N}(100)$$

$$c\% = \frac{\sum f}{N}(100)$$

EXAMPLE 5

Looking at Interval/Ratio Variables in Superfund Site Data

Table 7 indicates that two states have one Superfund site, one state has one site, and so on.

For one Superfund site: $\% = \frac{2 \text{ states}}{51}(100) = 3.9$ Therefore, 3.9% of all states have one Superfund site.

For two Superfund sites: $\% = \frac{1 \text{ state}}{51}(100) = 2.0$ Therefore, 2.0% of all states have two Superfund sites.

For three Superfund sites: $\% = \frac{1 \text{ state}}{51}(100) = 2.0$ Therefore, 2.0% of all states have three Superfund sites.

This process is repeated for each row in the table, so that each row consists of a value, frequency, percent, valid percent, and cumulative percent. Cumulative percents are now discussed.

The Cumulative Percent column is particularly useful for tables with interval/ratio variables. For example, if we want to know how many states have less than five Superfund sites, we look at the first column of Table 7 and identify those rows in the table that indicate 0–4 sites. When we move across the table, we find that the cumulative percent for four sites is 11.8%. This means that 11.8% of all states have four or fewer Superfund sites.

The Cumulative Percent column is best thought of as a running total of the Valid Percent column. It is important to realize, however, that the calculation of cumulative percents cannot be accomplished by adding up the valid percents. The reason for this is that it is possible that the rounding error in each valid percent equation could be

TABLE 7 *Frequency of Superfund Sites*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	1	2	3.9	3.9	3.9
	2	1	2.0	2.0	5.9
	3	1	2.0	2.0	7.8
	4	2	3.9	3.9	11.8
	8	1	2.0	2.0	13.7
	9	1	2.0	2.0	15.7
	11	1	2.0	2.0	17.6
	12	2	3.9	3.9	21.6
	13	2	3.9	3.9	25.5
	14	3	5.9	5.9	31.4
	15	2	3.9	3.9	35.3
	16	2	3.9	3.9	39.2
	17	1	2.0	2.0	41.2
	18	2	3.9	3.9	45.1
	20	4	7.8	7.8	52.9
	21	1	2.0	2.0	54.9
	22	2	3.9	3.9	58.8
	23	3	5.9	5.9	64.7
	29	1	2.0	2.0	66.7
	33	1	2.0	2.0	68.6
	34	2	3.9	3.9	72.5
	36	1	2.0	2.0	74.5
	40	1	2.0	2.0	76.5
	44	1	2.0	2.0	78.4
	45	1	2.0	2.0	80.4
	46	1	2.0	2.0	82.4
	51	1	2.0	2.0	84.3
	59	1	2.0	2.0	86.3
	65	1	2.0	2.0	88.2
	72	1	2.0	2.0	90.2
	84	1	2.0	2.0	92.2
	107	1	2.0	2.0	94.1
	110	1	2.0	2.0	96.1
	123	1	2.0	2.0	98.0
	140	1	2.0	2.0	100.0
Total		51	100.0	100.0	



AP Images/The News Tribune, Peter Haley

compounded when they are summed. The way to get around the possibility of compounded rounding error is to base the cumulative percent on cumulative frequencies. Using our example of states with 0–4 Superfund sites, we find that two states have 0 sites, one state has one site, one state has two sites, and two states have four sites. Therefore:

$$c\% = \frac{2 + 1 + 1 + 2 \text{ states}}{51}(100) = 11.8 \quad \text{Therefore, 11.8\% of all states have four or fewer Superfund sites.}$$

If we want to calculate how many states have 12 or fewer sites, our equation is as follows:

$$c\% = \frac{2 + 1 + 1 + 2 + 1 + 1 + 1 + 2 \text{ states}}{51}(100) = 21.6 \quad \text{Therefore, 21.6\% of all states have twelve or fewer Superfund sites.}$$

Now You Try It 4

Using the preliminary data from your study of integration into campus social life, construct a frequency table for the variable Time on Campus. Be sure to include value labels (in this case the actual values), frequencies, percents, valid percents, and cumulative percents.

RESPONDENT #	SEX	CLASS STANDING	TIME ON CAMPUS
1	Male	Freshman	90
2	Male	Junior	75
3	Female	Junior	60
4	Female	Sophomore	85
5	Female	Freshman	95
6	Male	Senior	50
7	Female	Junior	70
8	Female	Senior	60
9	Male	Sophomore	90
10	Female	Junior	70

*Answers at end of chapter.

Constructing Frequency Tables from Raw Data

Constructing frequency tables is a relatively simple process, but it requires careful attention to detail. Imagine that you walk into a typical classroom on a college campus and ask each student to tell you his or her age. You get the following results:

17, 18, 18, 19, 19, 19, 19, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 22, 24, 25

The following sections demonstrate how these data are organized into a frequency table.

The Frequency Column. Taking these 20 cases of raw data and organizing them into a frequency table begins with making a table with two columns, one that lists the attributes of our variable (the ages that respondents indicated) and one that lists the frequency (f) that each age occurred (Table 8).

The Percent Column. To calculate the Percent column, we divide each frequency by N to obtain the proportion. Then we multiply the proportion by 100 to obtain the percent.

$$17: \% = \frac{1}{20}(100) = 5.0\%$$

$$18: \% = \frac{2}{20}(100) = 10.0\%$$

$$19: \% = \frac{4}{20}(100) = 20.0\%$$

$$20: \% = \frac{7}{20}(100) = 35.0\%$$

$$21: \% = \frac{3}{20}(100) = 15.0\%$$

TABLE 8 *Age of Respondent*

	FREQUENCY (f)
17	1
18	2
19	4
20	7
21	3
22	1
24	1
25	1
Total (N)	20

TABLE 9 *Age of Respondent*

	FREQUENCY	PERCENT
17	1	5.0
18	2	10.0
19	4	20.0
20	7	35.0
21	3	15.0
22	1	5.0
24	1	5.0
25	1	5.0
Total	20	100.0

$$22: \% = \frac{1}{20}(100) = 5.0\%$$

$$24: \% = \frac{1}{20}(100) = 5.0\%$$

$$25: \% = \frac{1}{20}(100) = 5.0\%$$

We can then add the percentages to the Percent column in Table 9.

The Cumulative Percent Column. Finally, we can create a Cumulative Percent column ($c\%$) using cumulative frequencies (cf). Remember, Cumulative Percent columns summarize data up to a given point in a frequency table. For example, we may want to know what percent of respondents are younger than 20. To calculate this, we need to find out how many respondents are younger than 20 (which, in this case, is another way of asking for the cumulative frequency for 17, 18, and 19-year-olds). We do this as follows:

$$c\% = \frac{cf}{N}(100)$$

$$17: c\% = \frac{1}{20}(100) = 5.0\%$$

$$18: c\% = \frac{1+2}{20}(100) = 15.0\%$$

$$19: c\% = \frac{1+2+4}{20}(100) = 35.0\%$$

$$20: c\% = \frac{1+2+4+7}{20}(100) = 70.0\%$$

FREQUENCY TABLES

$$21: c\% = \frac{1 + 2 + 4 + 7 + 3}{20}(100) = 85.0\%$$

$$22: c\% = \frac{1 + 2 + 4 + 7 + 3 + 1}{20}(100) = 90.0\%$$

$$24: c\% = \frac{1 + 2 + 4 + 7 + 3 + 1 + 1}{20}(100) = 95.0\%$$

$$25: c\% = \frac{1 + 2 + 4 + 7 + 3 + 1 + 1 + 1}{20}(100) = 100.0\%$$

These values can now be added as another column in Table 10.

Getting back to our earlier question, because there are one 17-year old, two 18-year olds, and four 19-year olds, the cumulative frequency is $1 + 2 + 4 = 7$. Therefore, the cumulative percent for 19 year-olds is 35.0%. We can now state that 35.0% of our sample is less than 20 years old.

Integrating Technology

In its annual *Human Development Report*, the United Nations gathers tremendous amounts of data from countries all over the world and makes these data available to the public via the Internet. Data can be viewed online or downloaded into Microsoft Excel files (which can then be saved in a variety of formats that most statistics software programs can read). These include data on literacy rates, health care, carbon dioxide emissions, gender discrimination, vital statistics (births, deaths), and income inequality, among many others.

Follow this link for the most recent report: <http://hdr.undp.org/en/>

One interesting exercise is to sort countries from lowest to highest by different variables to see what other countries have similar characteristics. While we in the United States tend to compare the U.S. to many western European countries, you may be surprised to see that in terms of income inequality, the U.S. is much more similar to many less developed and “third world” nations.

TABLE 10 *Age of Respondent*

	FREQUENCY	PERCENT	CUMULATIVE PERCENT
17	1	5.0	5.0
18	2	10.0	15.0
19	4	20.0	35.0
20	7	35.0	70.0
21	3	15.0	85.0
22	1	5.0	90.0
24	1	5.0	95.0
25	1	5.0	100.0
Total	20	100.0	

EXAMPLE 6**Student Satisfaction with the College Environment**

Here is another example. Now suppose that for a class project, you are asked to survey students on their overall satisfaction with the overall quality of their college environment. You sample 500 students, asking them if they are “very satisfied,” “somewhat satisfied,” or “not satisfied.” You are now faced with the task of sifting through 500 sheets of paper with your respondents’ responses in an attempt to figure out how they tend to feel about their college experience. This is a good time to consider constructing a frequency table.

To construct a frequency table, you must tally the responses for each of the three attributes. Suppose you find that 342 respondents are “very satisfied,” 116 are “somewhat satisfied,” and 42 are “not satisfied.” You can now summarize your responses in a table format like the one in Table 11.

The next task is to display these data in the form of percents instead of frequencies. For example, if someone asked how satisfied students are with their college experience, you could answer that 342 out of 500 are very satisfied. This kind of answer, however, is not easy to interpret. It makes more sense to state our response as a percentage because it summarizes our findings by putting them into a format that is easy to understand and communicate.

As we did earlier, percentages for all of our responses are calculated using the formula

$$\% = \frac{f}{N}(100)$$

where f is equal to the frequency of a given attribute and N is the number of cases.

$$\text{Very satisfied: } \% = \frac{342}{500}(100) = 68.4$$

$$\text{Somewhat satisfied: } \% = \frac{116}{500}(100) = 23.2$$

$$\text{Not satisfied: } \% = \frac{42}{500}(100) = 8.4$$

TABLE 11 *How Satisfied are You with Your College Experience?*

	FREQUENCY (f)
Very satisfied	342
Somewhat satisfied	116
Not satisfied	42
Total	500

FREQUENCY TABLES

Often the percentages that we calculate do not add up to 100.0% like they do in this case. This is due to rounding error. With our percentages added, it looks like Table 12.

Finally, we can calculate the cumulative percent ($c\%$). Remember that cumulative percents are like running totals as we move down the list of attributes. For example, if we count all the respondents who are “very satisfied,” we are up to 68.4% of all respondents.

$$c\% = \frac{342}{500}(100) = 68.4$$

If we want to include all the respondents who are either “very satisfied” or “somewhat satisfied,” we have to add up the frequencies for each of the two attributes.

$$c\% = \frac{342 + 116}{500}(100) = 91.6$$

We can now state that 91.6% of all respondents are either “Very Satisfied” or “Somewhat Satisfied” with their college experience.

TABLE 12 *How Satisfied are You with Your College Experience?*

	FREQUENCY (f)	PERCENT (%)
Very satisfied	342	68.4
Somewhat satisfied	116	23.2
Not satisfied	42	8.4
Total	500	100.0

TABLE 13 *How Satisfied are You with Your College Experience?*

	FREQUENCY (f)	PERCENT (%)	CUMULATIVE % ($c\%$)
Very satisfied	342	68.4	68.4
Somewhat satisfied	116	23.2	91.6
Not satisfied	42	8.4	100.0
Total	500	100.0	

TABLE 14 *How Satisfied Are You with Your College Experience?*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Very satisfied	342	68.4	68.4	68.4
	Somewhat satisfied	116	23.2	23.2	91.6
	Not satisfied	42	8.4	8.4	100.0
	Total	500	100.0	100.0	

And if we want to include all the respondents who answered either “very satisfied,” “somewhat satisfied,” or “not satisfied” (which is everyone in the table), our equation looks like this:

$$c\% = \frac{342 + 116 + 42}{500}(100) = 100.0$$

Of course this is 100% of all the cases. It is tempting for many students to calculate the cumulative percent ($c\%$) by adding percentages from the Percent column, for example, $68.4 + 23.2\%$. However, this is risky because of the potential for rounding error in each percent to add up to a more significant rounding error when we combine the percentages. When the Cumulative Percent column is added, it looks like Table 13.

It is common for social scientists to use statistical software packages to enter and analyze their data. A common one is called SPSS for Windows. SPSS stands for “Statistical Program for the Social Sciences.” It is used to generate many of the tables and charts presented in this text. Table 14 is based on the same data as the one above and was generated using SPSS for Windows.

The table is neat, orderly, and easily interpreted. Best of all, many statistical software packages automatically generate these tables with just a few clicks of the mouse. The chart in Figure 7 provides the same information as the table, but does so in a way that emphasizes the trend that most students are very satisfied.

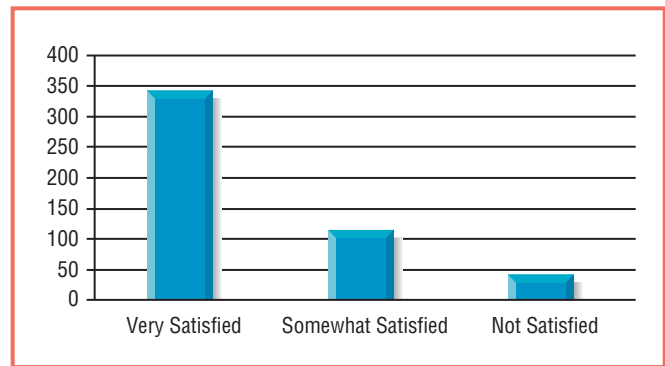


Figure 7 How Satisfied Are You with Your College Experience?

Statistical Uses and Misuses

Beware of “mindblowing” statistics! Newspapers, magazines, and other media outlets often report on rapid change in society. For example, we may read in the local paper that our community’s murder rate has jumped by 50% relative to a year ago. This may or may not be cause for concern. If the number of murders at this time last year was two, then a 50% increase means that our community has had three murders this year. An increase from two to three is not much of an increase (at least in most cities) and to report that as a 50% increase, while statistically true, is to mislead the consumers of these statistics by making them think the problem is bigger than it really is.

Another problem that can arise with percentages is related to the social and geographic specificity of statistics. Often, statistics that are intended to represent an entire society really only represent a portion of the population. For example, we may read that the number of murders so far this year is double the

number of murders from this time last year. This does not, however, mean that everyone is at twice the risk of being murdered. It may be that a vast majority of all murders take place in cities with populations of 250,000 or more. (I should state that this is only a hypothetical example and that I do not actually know where most murders take place.) It is therefore possible that the overall increase in urban murders is rising, and causing the nationwide murder rate to rise, while other types of communities are actually seeing declines in the murder rate.

The same could be true with any kind of social phenomena: child abductions, runaways, rapes, etc. It is important to know what kinds of questions to ask when we hear these “mind-blowing” statistics. What are the frequencies upon which these statistics are based? To what populations can these statistics be generalized? And maybe most importantly, who has an interest in reporting these kinds of statistics?

TABLE 15 *Ever Work as Long as One Year?*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	YES	1330	29.5	87.7	87.7
	NO	187	4.1	12.3	100.0
	Total	1517	33.6	100.0	
Missing	NAP	2988	66.3		
	NA	5	.1		
	Total	2993	66.4		
Total		4510	100.0		

Missing cases Cases in a frequency table that, for whatever reason, do not provide data.

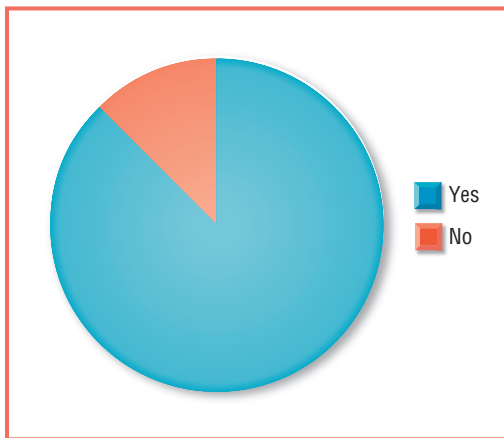


Figure 8 Ever Work as Long as One Year?

Frequency Tables and Missing Cases. As you can see by comparing these two tables, the one generated by SPSS for Windows provides an additional column, the “Valid Percent” column. The difference between “Percent” and “Valid Percent” is important. While Percent columns are based on the total number of cases in the data, Valid Percent columns are based on only the cases that provided data for the variable being analyzed. In other words, the value of *N* is often different for percent and valid percent. In the previous example, they are the same because all 500 respondents provided data; however, as Table 15 shows, this is not always the case. Figure 8 shows the data from Table 15 in the form of a pie chart. Notice that missing cases are not included in the pie chart and would only detract from the pattern in the data.

Table 15 is based on data from the 2006 General Social Survey. As you can see, the values of the “Percent” column differ a great deal from the values in the “Valid Percent” column. In fact, it is the difference between claiming that 29.5% of the sample has ever worked as long as a year and claiming that 87.7% of the sample has ever worked as long as a year—a significant discrepancy to say the least! The Percent column values are based on an *N* of 4,510 (the total number of people included in the sample). All of these cases are included even though 2,993 of them did not provide any data. The formula for this percent is:

$$\% = \frac{1330}{4510}(100) = 29.5$$

The Valid Percent column values are based on an *N* of 1,517, which includes only those respondents who provided data for the table. The formula for the valid percent is:

$$\text{Valid \%} = \frac{1330}{1517}(100) = 87.7$$

It is preferable to use the Valid Percent over the Percent because missing cases really don’t tell us anything. Would you feel comfortable claiming that 29.5% of the sample

has ever worked as long as one year when 2,993 (66.4%) of those people did not provide any data at all? Of course not! That is why we use the valid percent.

You can see that Table 15 indicates that 2,993 cases are missing. Cases may be missing for a variety of reasons. For example, in this table, 2,988 respondents indicated that the question was not applicable (NAP) to them (we don't know why) and that another 5 respondents did not answer (NA) the question (again, we do not know why).

Table 16 is a frequency table for a variable that describes respondents' political views. The first column lists the attributes of the variable "Think of Self as Liberal or Conservative." The frequency column shows the number of respondents who answered "extremely liberal," "liberal," and so on. The percent column shows the percentage out of all 1,500 respondents who were presented with the questionnaire. The Valid Percent column shows only the percent of responses based on the number of valid responses ($N = 1,443$). In other words, 1,500 respondents were included in the study, yet only 1,443 respondents provided valid responses. Of the 57 respondents who did not provide data, 48 did not know what to answer (DK) and nine indicated that the questionnaire item was not applicable to them (NA). Because these 57 respondents provide no data, it is a good idea to leave them out of the analysis. Finally, the Cumulative Percent column, which is based on only those valid cases providing data, represents a "running total" of the data from one row to the next.

Generally, missing cases are left out of all analyses. Many students see this as problematic, but it is important to remember that the data is *missing*. For whatever reason, a respondent chose not to give us that information or the question did not apply to them. Therefore, it is common practice to exclude missing cases. The Valid Percent and Cumulative Percent columns in the frequency table are calculated on

TABLE 16 *Think of Self as Liberal or Conservative*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Extremely liberal	30	2.0	2.1	2.1
	Liberal	163	10.9	11.3	13.4
	Slightly liberal	193	12.9	13.4	26.7
	Moderate	527	35.1	36.5	63.3
	Slightly conservative	248	16.5	17.2	80.5
	Conservative	241	16.1	16.7	97.2
	Extremely conservative	41	2.7	2.8	100.0
Missing	Total	1443	96.2	100.0	
	DK	48	3.2		
	NA	9	.6		
	Total	57	3.8		
Total		1500	100.0		

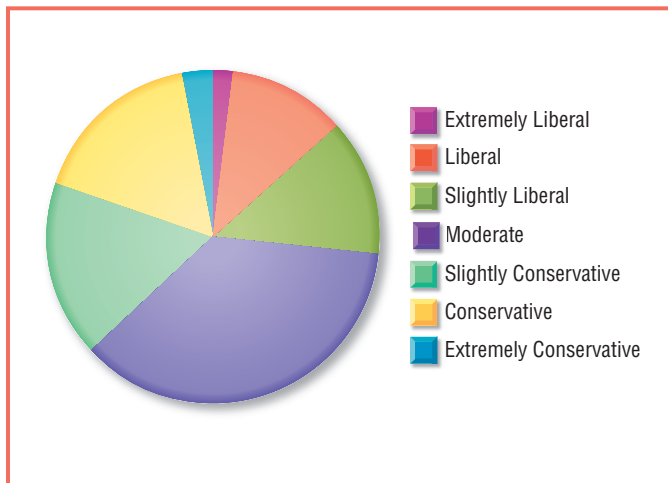


Figure 9 Think of Self as Liberal or Conservative

the basis of no missing cases. In Table 16, you can see that the response “extremely liberal” has a percent of 2.0 and a valid percent of 1. We see that 30 respondents are extremely liberal. This represents 2.0% of the 1500 respondents polled and 2.1% of the 1,443 respondents who answered the question. The difference is this: The percent is based on 1,500 cases and the valid percent is based on 1,443 cases. If you feel that it is important to include missing cases, read the Percent column instead of the Valid Percent column. A graphic representation (in this case it is a pie chart) of Table 16 is shown in Figure 9.

Eye on the Applied

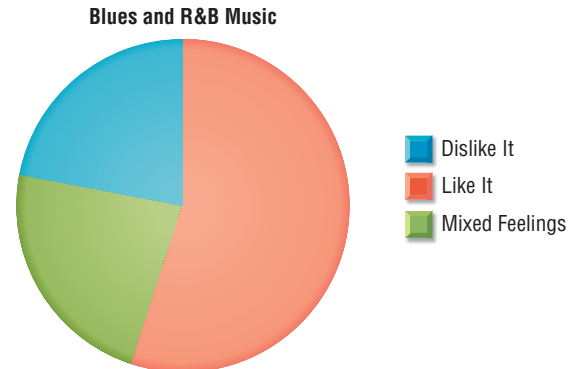
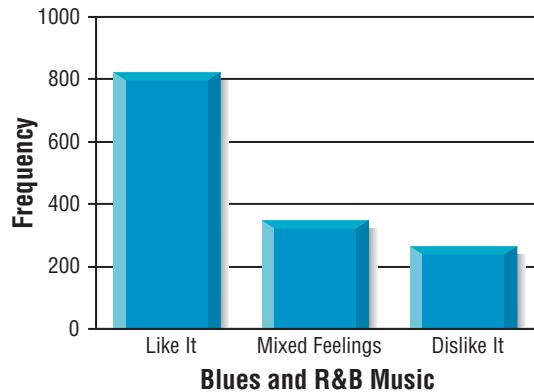
Describing and Summarizing Data

Frequency tables may be the most commonly used method of summarizing and communicating large amounts of data. They are much more common than you might think. A quick test of this can be done using a *USA Today* newspaper or a *Time* or *Newsweek* magazine. Take a few minutes and flip through one of these publications, making note of the number of pie charts and bar graphs you see. Each pie chart and bar graph is based on a frequency table. A wide array of computer software programs allow users to quickly change the format in which data is presented from a table, which is understood by those trained to read tables, to graphic forms, which most people can easily understand. The example below shows the same data in three different formats.

Blues and R&B Music

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Like It	823	54.9	57.4	57.4
	Mixed Feelings	348	23.2	24.3	81.7
	Dislike It	263	17.5	18.3	100.0
	Total	1434	95.6	100.0	
Missing	System	66	4.4		
Total		1500	100.0		

FREQUENCY TABLES



As you can see, a significant amount of data is lost when presented in graphic form as opposed to a table.

Chapter Summary

This chapter is all about how to organize data so that we can more easily summarize results, identify trends, and communicate results with other scientists. Two of the most popular ways to summarize data are in frequency tables and cross-tabulation tables. Frequency tables show the distribution of cases across the attributes of a single variable whereas cross-tabulation tables show the distribution of cases across two variables. These tables make

extensive use of both frequencies and percentages (percent, valid percent, cumulative percent, column percent, row percent, and total percent).

In addition to tables, this chapter also emphasizes the presentation of data in more visual forms such as charts (pie charts, bar charts). As a general rule, pie charts are used for nominal variables, bar charts are used for either nominal or ordinal variables.

Exercises

Use the table below to answer the following questions:

Religious Preference

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Protestant	953	63.5	63.9	63.9
	Catholic	333	22.2	22.3	86.2
	Jewish	31	2.1	2.1	88.3
	None	140	9.3	9.4	97.7
	Other	35	2.3	2.3	100.0
	Total	1492	99.5	100.0	
Missing	DK	1	.1		
	NA	7	.5		
	Total	8	.5		
Total		1500	100.0		

FREQUENCY TABLES

1. How many respondents provided data for the table?
2. How many respondents did not provide data?
3. What percent of respondents is Protestant?
4. How many of the respondents are Protestant?
5. Is the variable nominal, ordinal, or interval/ratio?
6. What proportion of respondents is Catholic?
7. What proportion of respondents is Jewish?

Use the table below to answer the following questions:

How Do You Get to Class?

Car	37
Public transportation	18
Walk	28
Bike	12
Other	20
Total	115

8. How many respondents provided data for this table?
9. What percent of respondents walk to class?
10. What percent of respondents either bike or walk?
11. What percent of respondents do not take public transportation?

Use the frequencies in the table below to fill in the percent, valid percent, and cumulative percent

columns. Then use the table to answer the following questions.

Think of Self as Liberal or Conservative

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Extremely Liberal	139			
	Liberal	524			
	Slightly Liberal	517			
	Moderate	1683			
	Slightly Conservative	618			
	Conservative	685			
	Extremely Conserva- tive	167			
	Total	4333			
Missing	DK	154			
	NA	23			
	Total	177			
Total		4510	100.0		

12. Is this variable nominal, ordinal, or interval/ratio?
13. How many respondents provided data for this table?
14. What percent of respondents consider themselves moderate (always use valid percent)?
15. What percent of respondents consider themselves more liberal than moderate?
16. What percent of respondents are either slightly conservative or slightly liberal?

Computer Exercises

Construct a frequency table using the data below. Then use the table to answer the following questions.

How Often Do You Use the Library?

Daily	Several times a week	Weekly	Daily	Several times a week
Several times a week	Daily	Several times a week	Monthly	Never
Weekly	Daily	Never	Several times a week	Several times a week
Daily	Several times a week	Several times a week	Several times a week	Daily
Never	Weekly	Several times a week	Monthly	Weekly

17. Is this variable nominal, ordinal, or interval/ratio?
 18. How many attributes does this variable have?
 19. What is N equal to?
 20. How many respondents use the library on a daily basis?
 21. What percent of respondents use the library several times a week?
 22. What percent of respondents use the library more often than monthly?
- Answer the following questions using the GSS2006 data provided.
23. How many respondents are married?
 24. What proportion of respondents are married?
 25. What percent of respondents are divorced?
 26. Of males, what proportion are separated?
 27. Of females, what percent are widowed?
 28. What is the ratio of married to unmarried respondents?

Key Terms

Cross-tabulation table
Cumulative percent
Data
Ecological data
Frequency

Frequency table
Individual data
Missing cases
Percent
Percentage

Proportion
Ratio
Valid percent
Value labels

Now You Try It Answers

1: 1. .736 2. 77.7% 3. 63.4% 4. 2.4
2:

Sex of Respondent

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Male	4	40.0	40.0	40.0
	Female	6	60.0	60.0	100.0
	Total	10	100.0	100.0	

3:

Class Standing of Respondent

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Freshman	2	20.0	20.0	20.0
	Sophomore	2	20.0	20.0	40.0
	Junior	4	40.0	40.0	80.0
	Senior	2	20.0	20.0	100.0
	Total	10	100.0	100.0	

4:

Percent of Time R Spends on Campus

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	50	1	10.0	10.0	10.0
	60	2	20.0	20.0	30.0
	70	2	20.0	20.0	50.0
	75	1	10.0	10.0	60.0
	85	1	10.0	10.0	70.0
	90	2	20.0	20.0	90.0
	95	1	10.0	10.0	100.0
	Total	10	100.0	100.0	

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